Haotian Jiang

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Education

University of Wisconsin–Madison, Madison, WI — Sept 2025 – Present

- **M.A. in Mathematics**, expected May 2027.
- **Selected Coursework:** Real Analysis II, Modern Algebra II, Numerical Analysis, Stochastic Processes, Mathematical Methods for Systems Biology, Methods for Applied Mathematics, Data-Driven Dynamical Systems, Stochastic Modeling and Prediction.
- **Seminar:** Manifold Learning and Dimension Reduction for Complex Data.

Southern University of Science and Technology (SUSTech), Shenzhen, China — Sept 2022 – June 2026

- **B.S. in Mathematics**.
- **Selected Coursework:** Real Analysis, Probability and Statistics, Ordinary and Partial Differential Equations, Complex Analysis, Numerical Solution of Partial Differential Equations, Operations Research and Optimization, Artificial Intelligence, Python Programming.
- **Seminar:** Optimal Transport.

Research Experience

Research Project with Prof. Xin (Maizie) Zhou — 2026 – Present

- Investigating computational methods for **cell–cell communication** in single-cell and spatial omics data.
- Exploring how **optimal transport** can be incorporated into cell–cell communication inference, with CellChat and related ligand–receptor-based methods used as reference frameworks.

Undergraduate Researcher, SUSTech — 2025 – 2026

Advisor: Prof. Zhen Zhang

- Conducted undergraduate thesis research on **single-cell dynamical trajectory inference based on unbalanced optimal transport (UOT)**.
- Reproduced and extended the **TIGON** framework for inferring cellular dynamics from time-course single-cell transcriptomic data.
- Investigated algorithmic modifications, including regularization strategies, and evaluated inferred gene-regulatory dynamics using quantitative metrics.
- Thesis received the **Outstanding Undergraduate Thesis Award, Class of 2026, SUSTech**.

Honors and Awards

Outstanding Undergraduate Thesis Award, Class of 2026, SUSTech — June 2026

Honorable Mention, Mathematical Contest in Modeling (MCM) — Jan 2025

Third Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Guangdong Division — Nov 2024

Student Cadre of Excellence, SUSTech — Jan 2024

Second-Class Scholarship for Freshmen, SUSTech — Sept 2022

Technical Skills

Programming: Python, MATLAB.

Scientific Computing / ML: PyTorch, NumPy, SciPy, Pandas, POT.

Computational Biology: Scanpy; single-cell RNA-seq analysis; trajectory inference; cell–cell communication analysis.

Tools: Git, LaTeX.

Additional Training and Activities

Oxford International Study Abroad Programme, UK – Artificial Intelligence and Machine Learning — August 2024